ACETONE

IDENTITY

C3H6O • 58.08 g/mol

IUPAC name: propan-2-one

STRUCTURE & FUNCTIONAL GROUPS

CH■ — C(=O) — CH■

Condensed structural formula

- Condensed structural formula: CH3–C(=O)–CH3
- A carbonyl group within the carbon chain
- Functional group: ketone

KEY GROUPS

Ketone (carbonyl, C=O)

IMPORTANT PROPERTIES

- Colorless, volatile liquid with a characteristic odor.
- Miscible with water and many organic solvents.
- Low boiling point (about 56 °C at 1 atm) makes it easy to evaporate and recover by distillation.

WHAT IS AMAZING ABOUT IT?

Its volatility is useful—and a safety concern

Acetone's relatively low boiling point helps it evaporate quickly and makes distillation practical. The same volatility creates flammable vapor, so ventilation and ignition control matter.

USES & VALUE

- Solvent in coatings, cleaning, and laboratory work
- Feedstock in chemical manufacturing
- Used in some nail-polish removers and formulations

RISKS & SAFETY

- Highly flammable liquid and vapor.
- May irritate eyes and cause dizziness at high vapor concentrations.
- Keep away from ignition sources; use appropriate ventilation and storage.

CHEMICAL ENGINEERING CONNECTION

Acetone recovery can use vapor–liquid equilibrium and distillation. Process design must balance energy use, solvent recovery, emissions control, and fire protection.

SELECTED REFERENCES

National Center for Biotechnology Information. (n.d.). PubChem compound summary for acetone (CID 180). https://pubchem.ncbi.nlm.nih.gov/compound/180
National Institute for Occupational Safety and Health. (n.d.). NIOSH pocket guide to chemical hazards: Acetone. Centers for Disease Control and Prevention. https://www.cdc.gov/niosh/npg/